

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Industry Problem Task

Duration: 2.5hrs

1. Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time. **(5 Marks)**

2. Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy. **(5 Marks)**

3. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance. **(5 Marks)**

4. Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments. **(5 Marks)**

5. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance. **(5 Marks)**

6. Apply Reinforcement Learning to dynamic pricing in e-commerce. Define the state space, action space, and reward function. Discuss challenges in balancing exploration and exploitation. **(5 Marks)**

7. Illustrate the use of Reinforcement Learning in smart city waste management systems. Identify the MDP components and discuss how continuous updates improve operational efficiency. **(5 Marks)**

8. Evaluate how Reinforcement Learning can be used in network bandwidth allocation in telecommunications. Define states, actions, and rewards, and examine how the system adapts to dynamic conditions. **(5 Marks)**

9. Analyze the application of Reinforcement Learning in portfolio management in finance. Discuss advantages over traditional models and challenges such as risk and delayed rewards. **(5 Marks)**

10. Justify the use of Reinforcement Learning in renewable energy management systems. Define the RL framework components and discuss the impact of environmental uncertainty on decision-making. **(5 Marks)**

Code of Conduct:

I T.chaitra(192425217)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T.chaitra

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

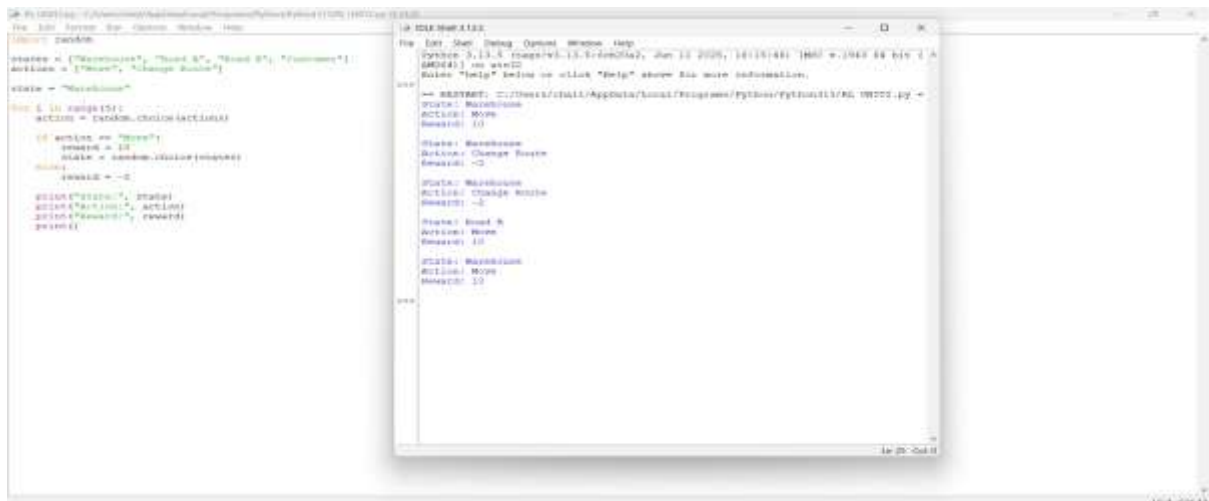
1. Reinforcement Learning for Logistics Delivery Route Optimization

Reinforcement Learning (RL) can optimize delivery routes by enabling a delivery vehicle (agent) to learn the most efficient path through continuous interaction with the environment. The objective is to reduce delivery time, fuel consumption, and operational costs.

RL Components:

- **States:** Current vehicle location, traffic condition, weather, fuel level, pending deliveries.
- **Actions:** Move to the next location, change route, stop for refueling, deliver package.
- **Rewards:** Positive reward for on-time deliveries and shorter routes; negative reward for delays, traffic congestion, and higher fuel consumption.
- **Policy:** A strategy that selects the best action for each state to maximize long-term rewards.

Continuous learning allows the agent to improve its routing decisions using previous experiences, making deliveries faster and more efficient over time.



2. Reinforcement Learning in Fraud Detection

Reinforcement Learning helps financial institutions identify fraudulent transactions by learning from previous transaction outcomes. The agent continuously improves its decision-making process as new transaction data becomes available.

RL Components:

- **State Space:** Transaction amount, customer location, account history, device information, transaction frequency.
- **Action Space:** Approve transaction, reject transaction, request additional verification.

- **Reward Function:** Positive reward for correctly identifying fraud and legitimate transactions; negative reward for false positives and missed fraud cases.

Exploration strategies allow the system to test different detection methods, helping it discover new fraud patterns and improve detection accuracy over time.

The image shows a Jupyter Notebook interface with a code cell and its corresponding terminal output.

Code Cell:

```
transactions = ["Normal", "Fraud"]

for i in range(5):
    state = random.choice(transactions)
    action = random.choice(["Approve", "Reject"])

    if state == "Fraud" and action == "Reject":
        reward = 10
    elif state == "Normal" and action == "Approve":
        reward = 10
    else:
        reward = -1

    print(state, action, reward)
```

Terminal Output:

```
Python 3.13.5 Image/Win32: 3.13.5:64bit0042, Jun 11 2025, 16:11:46) [MSC v.1963 64 bit (AMD64)] on win32
Enter "help" alone or click "Help" above for more information.

>>> RESTART: c:/Users/cha1t/AppData/Local/Programs/Python/Python313/UNT-2.py >>>
Fraud Reject 10
Normal Approve 10
Normal Reject -1
Normal Approve 10
Normal Approve 10
```

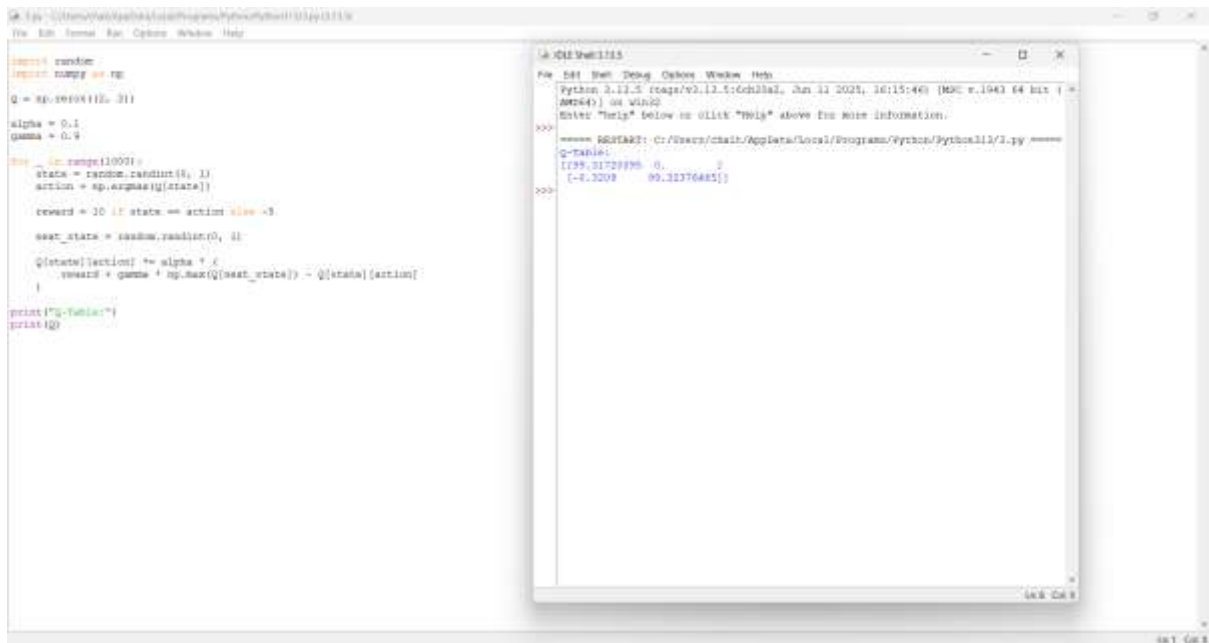
3. Reinforcement Learning in Streaming Platform Recommendation

Streaming platforms use Reinforcement Learning to recommend movies, music, or TV shows based on user preferences and viewing history. The recommendation system learns from user interactions to improve future suggestions.

Key Concepts:

- The recommendation system acts as the **agent**.
- User preferences represent the **environment**.
- Positive rewards are received when users watch, like, or rate recommended content.
- Negative rewards occur when users skip or dislike recommendations.

Delayed rewards occur because users may interact with recommendations later. Continuous user feedback and changing preferences help the system learn and provide more personalized recommendations.



4. Reinforcement Learning in Predictive Maintenance

Predictive maintenance uses Reinforcement Learning to determine the best time to maintain industrial equipment based on real-time sensor data instead of fixed maintenance schedules.

Advantages over Rule-Based Systems:

- Learns from machine operating conditions.
- Adapts to changing environments.
- Reduces maintenance costs and equipment downtime.
- Improves equipment lifespan.

Risks and Reliability Concerns:

- Incorrect decisions may lead to equipment failure.
- Requires large amounts of quality data.
- Safety-critical industries require extensive testing before deployment.

Overall, RL provides more flexible and intelligent maintenance planning than traditional rule-based methods.

The image shows a Jupyter Notebook on the left and a terminal window on the right. The Jupyter Notebook contains the following code:

```

import numpy as np

states=["empty","full"]
actions=["continue","stop"]

Q=np.zeros((2,2))

alpha=0.1
gamma=0.9

def s_in_range(100):
    state=round(random()*100,1)
    action=np.argmax(Q[state])
    if state==1 and action==1:
        reward=10
    elif state==0 and action==0:
        reward=0
    else:
        reward=-5
    next_state=round(random()*100,1)
    Q[state][action]=alpha*(reward+gamma*np.max(Q[next_state])-Q[state][action])
    print(Q)

```

The terminal window shows the output of the code, displaying the Q-table values and the action taken for a given state.

```

python 3.12.5 (tags/v3.12.5:cd00000, Jan 11 2025, 16:15:46) [AMD v.1944] 64 bit | >
AND641 | on win32
Enter "help" below or click "help" above for more information.

===== RESTART: C:/Users/cha1t/AppData/Local/Programs/Python/Python312/python.exe =====
[[[-0.8, 77.37762566]]]

```

5. Reinforcement Learning in Transportation Systems

Reinforcement Learning improves transportation systems by optimizing bus schedules and route allocation based on real-time traffic and passenger demand.

RL Components:

- **States:** Bus locations, passenger demand, traffic congestion, bus occupancy.
- **Actions:** Change routes, adjust schedules, increase or decrease bus frequency.
- **Rewards:** Reduced passenger waiting time, lower fuel consumption, improved service quality.

Real-time traffic information enables the RL agent to continuously adapt its decisions, improving transportation efficiency and passenger satisfaction.

The image shows a Jupyter Notebook on the left and a terminal window on the right. The Jupyter Notebook contains the following code:

```

# Traffic = 1 (High) or 0 (Low)
# Passenger = 1 (High) or 0 (Low)
# Action = 1 (Change Route) or 0 (Keep Bus)

# Traffic = 1 (High) and Passenger = 1 (High)
# Action = 1 (Change Route)
# Reward = 10

# Traffic = 1 (High) and Passenger = 0 (Low)
# Action = 0 (Keep Bus)
# Reward = 5

# Traffic = 0 (Low) and Passenger = 1 (High)
# Action = 1 (Change Route)
# Reward = 5

# Traffic = 0 (Low) and Passenger = 0 (Low)
# Action = 0 (Keep Bus)
# Reward = 0

# Action = 1 (Change Route)
# Reward = 10

# Action = 0 (Keep Bus)
# Reward = 0

print("Action:", action)
print("Reward:", reward)

```

The terminal window shows the output of the code, displaying the current state, action, and reward.

```

python 3.12.5 (tags/v3.12.5:cd00000, Jan 11 2025, 16:15:46) [AMD v.1944] 64 bit | >
AND641 | on win32
Enter "help" below or click "help" above for more information.

===== RESTART: C:/Users/cha1t/AppData/Local/Programs/Python/Python312/python.exe =====
Current State:
Traffic: 1
Passenger: 1
Action: 1
Reward: 10

```

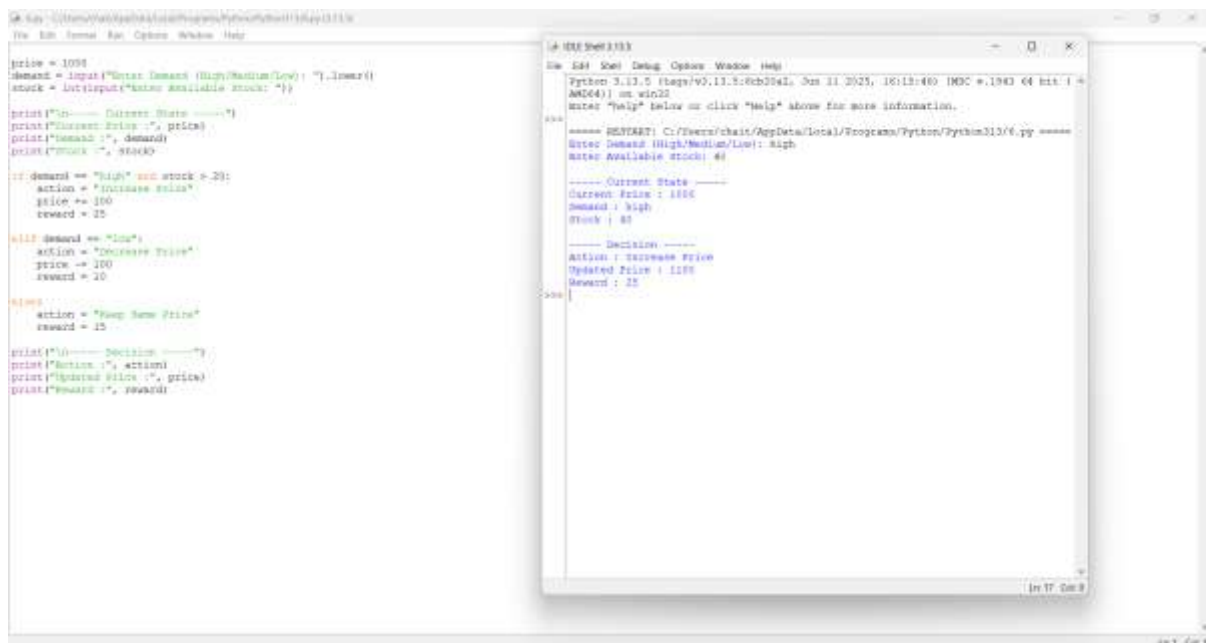
6. Reinforcement Learning in Dynamic Pricing

Reinforcement Learning helps e-commerce platforms automatically adjust product prices according to demand, competition, and customer behavior.

RL Components:

- **State Space:** Product demand, competitor prices, inventory level, customer behavior, seasonal trends.
- **Action Space:** Increase price, decrease price, maintain current price.
- **Reward Function:** Higher sales, increased profit, improved customer satisfaction.

The main challenge is balancing **exploration** (trying new prices) and **exploitation** (using successful pricing strategies). A proper balance maximizes long-term profits.



The image shows a Python script in a text editor and its execution output in a terminal window. The script implements a simple reinforcement learning model for dynamic pricing. It starts with a price of 1000 and a stock of 100. The user is prompted to enter demand (High/Medium/Low). The script then calculates the current state (price, demand, stock) and prompts the user to enter an action (Increase/Decrease/Same). The action is then executed, and the new price, demand, and stock are displayed. The script also calculates a reward based on the action and the current state.

```
price = 1000
demand = input("Enter Demand (High/Medium/Low): ")
stock = input("Enter Available Stock: ")

print("Current State:")
print("Current Price: ", price)
print("Demand: ", demand)
print("Stock: ", stock)

if demand == "high" and stock > 20:
    action = "Increase Price"
    price += 100
    reward = 25
elif demand == "low":
    action = "Decrease Price"
    price -= 100
    reward = 10
else:
    action = "Keep Same Price"
    reward = 15

print("Decision:")
print("Action: ", action)
print("Updated Price: ", price)
print("Reward: ", reward)
```

Terminal Output:

```
Python 3.11.5 (tags/v0.11.5:8c3b0e1, Oct 11 2023, 18:13:40) [AMD64] on win32
Enter Demand (High/Medium/Low): high
Enter Available Stock: 80

Current State:
Current Price : 1000
Demand : high
Stock : 80

Decision:
Action : Increase Price
Updated Price : 1100
Reward : 25
```

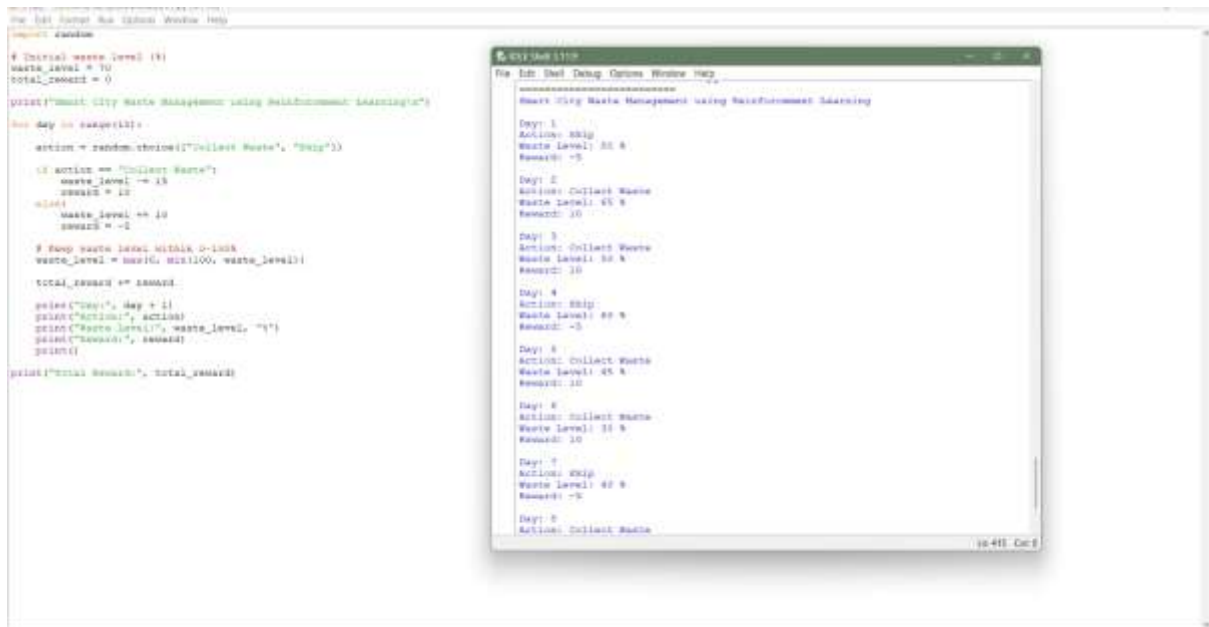
7. Reinforcement Learning in Smart City Waste Management

Reinforcement Learning optimizes waste collection by selecting efficient routes and schedules based on real-time information from smart bins.

MDP Components:

- **States:** Bin fill level, vehicle location, traffic conditions.
- **Actions:** Dispatch collection vehicle, choose collection route, postpone collection.
- **Rewards:** Reduced fuel consumption, timely waste collection, lower operational costs.
- **Policy:** Select the most efficient waste collection strategy.

Continuous sensor updates help the system adapt to changing waste levels, improving operational efficiency and reducing unnecessary collection trips.



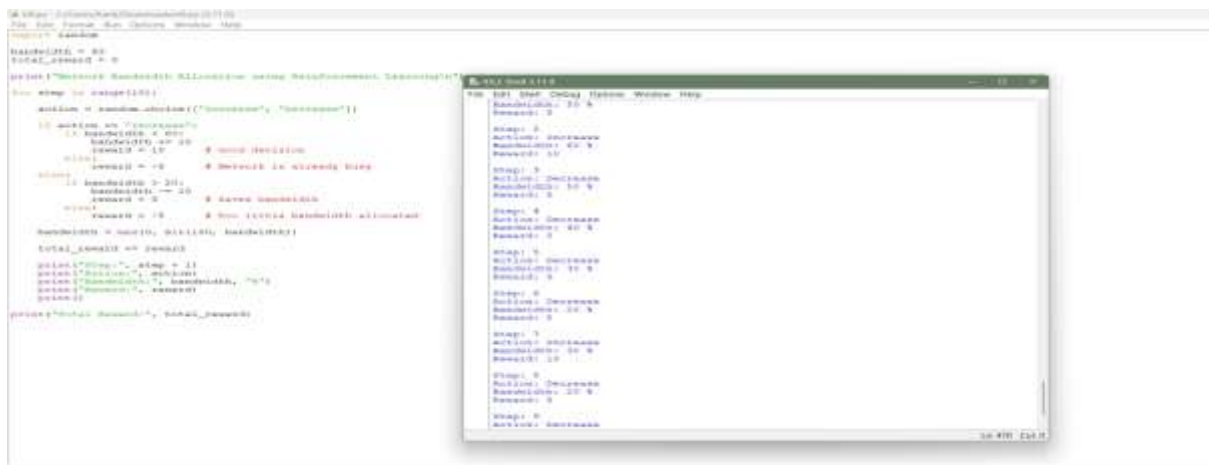
8. Reinforcement Learning in Network Bandwidth Allocation

Reinforcement Learning dynamically allocates network bandwidth according to changing user demand and traffic conditions.

RL Components:

- **States:** Current bandwidth usage, network congestion, number of connected users.
- **Actions:** Increase bandwidth, decrease bandwidth, redistribute bandwidth.
- **Rewards:** Higher network throughput, lower latency, improved Quality of Service (QoS).

The RL agent continuously learns from network conditions and adapts resource allocation, ensuring efficient communication even during heavy traffic.



9. Reinforcement Learning in Portfolio Management

Reinforcement Learning helps investors optimize investment decisions by learning from market performance and adjusting portfolio allocations over time.

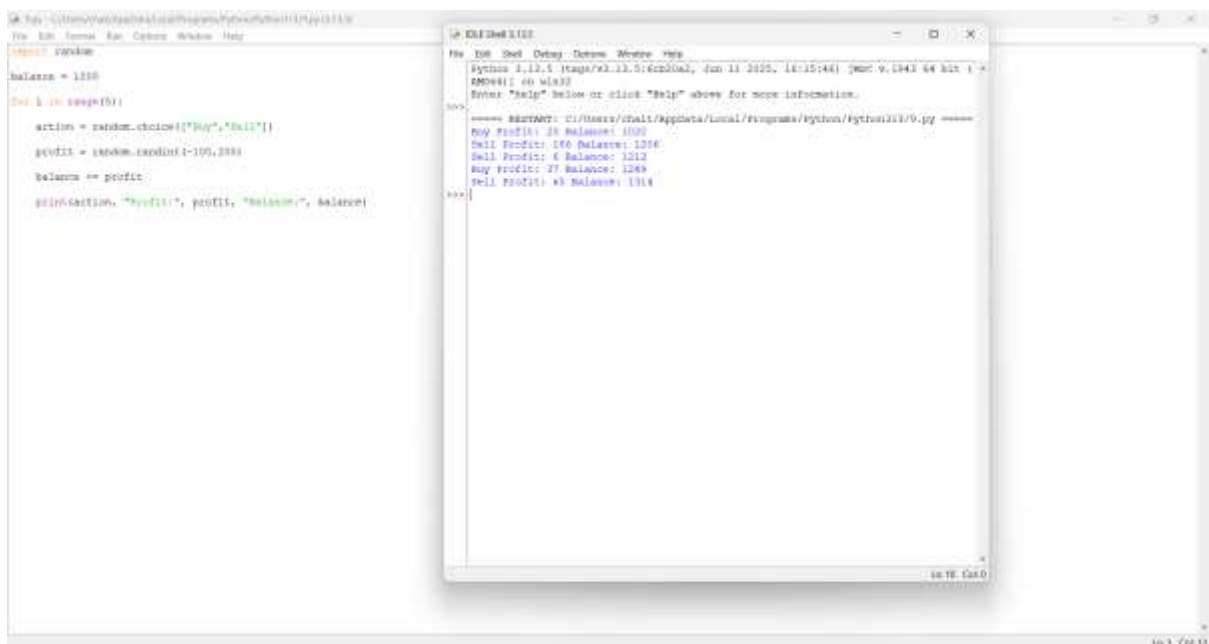
Advantages:

- Learns from changing market conditions.
- Adapts investment strategies automatically.
- Optimizes long-term returns.
- Handles complex financial environments.

Challenges:

- Delayed rewards because profits are realized over time.
- High market uncertainty and financial risk.
- Requires large historical datasets for effective learning.

Despite these challenges, RL offers greater adaptability than traditional investment methods.



The image shows a screenshot of a Python script being executed in a terminal window. The script is located at `C:\Users\stait\AppData\Local\Programs\Python\Python311\Scripts\python.exe` and is named `rl_portfolio.py`. The script's content is as follows:

```
import random

balance = 1000

def simulate():
    action = random.choice(["buy", "sell"])
    profit = random.randint(-100, 200)
    balance += profit
    print(action, "profit:", profit, "balance:", balance)
```

The terminal output shows the execution of the script, displaying the results of the simulation:

```
python rl_portfolio.py
=====
Buy Profit: 25 Balance: 1025
Sell Profit: 150 Balance: 1175
Buy Profit: 5 Balance: 1180
Sell Profit: 27 Balance: 1207
Buy Profit: 45 Balance: 1252
```

10. Reinforcement Learning in Renewable Energy Management

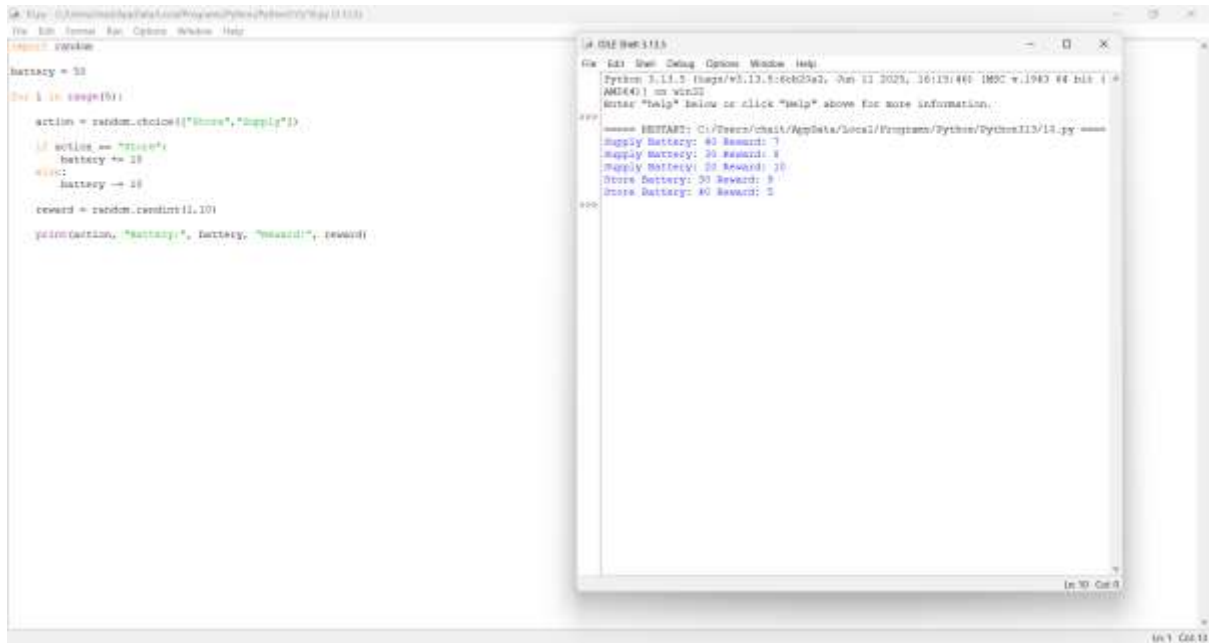
Reinforcement Learning improves renewable energy management by optimizing energy generation, storage, and distribution in smart grids.

RL Framework Components:

- **States:** Energy demand, battery charge level, weather conditions, solar and wind power generation.
- **Actions:** Store energy, distribute electricity, purchase power from the grid.

- **Rewards:** Reduced energy cost, efficient power utilization, stable electricity supply.
- **Policy:** Choose actions that maximize long-term energy efficiency.

Environmental uncertainty, such as varying sunlight and wind speed, affects decision-making. RL continuously learns from these changes, enabling better energy management and improving the reliability of renewable energy systems.



The image shows a Python script in a text editor and its execution output in a terminal window.

Python Script (Left Window):

```

battery = 10
for i in range(5):
    action = random.choice(["Store", "Supply"])
    if action == "Supply":
        battery += 10
    else:
        battery -= 10
    reward = random.randint(1, 10)
    print(action, "Battery:", battery, "Reward:", reward)

```

Terminal Output (Right Window):

```

Python 3.11.5 (tags/v3.11.5:fc86282, Jun 11 2023, 16:11:40) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "help" above for more information.

==== RESTART: C:/Users/cheit/AppData/Local/Programs/Python/Python113/13.py ====
Supply Battery: 10 Reward: 7
Supply Battery: 20 Reward: 8
Supply Battery: 30 Reward: 10
Store Battery: 20 Reward: 9
Store Battery: 10 Reward: 5

```